

TABELLE DEL PAPER

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September 2026

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1 FULL PANEL

Table 1: Full panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0023 (0.0039)	−0.0013 (0.0041)	−0.0004 (0.0013)	−0.0001 (0.0010)	−0.00003 (0.0011)	−0.0001 (0.0003)
Asymptotic p	[0.566]	[0.750]	[0.748]	[0.924]	[0.974]	[0.857]
Bootstrap p	[0.686]	[0.851]	[0.740]	[0.931]	[0.978]	[0.860]
$EP_{dt} \times b_p$	−0.0044* (0.0022)	−0.0048* (0.0029)	0.0001 (0.0010)	−0.0009 (0.0006)	−0.0006 (0.0007)	−0.0001 (0.0004)
Asymptotic p	[0.052]	[0.096]	[0.903]	[0.153]	[0.409]	[0.697]
Bootstrap p	[0.185]	[0.391]	[0.937]	[0.177]	[0.443]	[0.734]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,313,047	21,313,037	21,519,511	21,313,047	21,313,037
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0019 (0.0037)	−0.0015 (0.0034)	−0.0004 (0.0019)	−0.0001 (0.0010)	−0.0001 (0.0010)	−0.0001 (0.0005)
Asymptotic p	[0.613]	[0.660]	[0.819]	[0.904]	[0.917]	[0.776]
Bootstrap p	[0.673]	[0.755]	[0.843]	[0.920]	[0.933]	[0.852]
$EP_{dt} \times b_p$	−0.0056** (0.0027)	−0.0070** (0.0030)	0.0004 (0.0011)	−0.0014** (0.0007)	−0.0014* (0.0008)	−0.00004 (0.0003)
Asymptotic p	[0.039]	[0.020]	[0.676]	[0.048]	[0.080]	[0.903]
Bootstrap p	[0.049]	[0.035]	[0.765]	[0.069]	[0.100]	[0.915]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,517,666	21,311,206	21,311,196	21,517,666	21,311,206	21,311,196
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963

Notes: Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv} = \ln x_{fpdt}^{val} - \ln x_{fpdt}^{qua}$). Quantity and unit-value regressions are estimated on the subset of observations with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

2 COLLAPSED PANEL

Table 2: Collapsed panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0046 (0.0069)	−0.0055 (0.0069)	0.0005 (0.0028)	0.0018 (0.0018)	0.0019 (0.0020)	−0.0001 (0.0007)
Asymptotic p	[0.508]	[0.426]	[0.854]	[0.316]	[0.360]	[0.868]
Bootstrap p	[0.649]	[0.525]	[0.851]	[0.390]	[0.433]	[0.867]
Permutation p	[0.597]			[0.160]		
$EP_{dt} \times b_p$	−0.0119*** (0.0029)	−0.0115* (0.0067)	0.0006 (0.0042)	0.0004 (0.0016)	−0.0004 (0.0019)	0.0009 (0.0005)
Asymptotic p	[0.000]	[0.087]	[0.880]	[0.825]	[0.837]	[0.076]
Bootstrap p	[0.072]	[0.250]	[0.947]	[0.857]	[0.886]	[0.169]
Permutation p	[0.278]			[0.817]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,681,023	3,672,709	3,672,709	3,681,023	3,672,709	3,672,709
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0043 (0.0043)	−0.0084* (0.0044)	0.0038 (0.0042)	0.0016 (0.0020)	0.0010 (0.0022)	0.0006 (0.0008)
Asymptotic p	[0.320]	[0.058]	[0.368]	[0.428]	[0.649]	[0.495]
Bootstrap p	[0.546]	[0.268]	[0.446]	[0.508]	[0.712]	[0.542]
Permutation p	[0.480]			[0.316]		
$EP_{dt} \times b_p$	−0.0113*** (0.0022)	−0.0097** (0.0045)	−0.0012 (0.0029)	−0.0002 (0.0014)	−0.0008 (0.0014)	0.0007 (0.0006)
Asymptotic p	[0.000]	[0.033]	[0.689]	[0.858]	[0.603]	[0.252]
Bootstrap p	[0.047]	[0.311]	[0.757]	[0.905]	[0.876]	[0.332]
Permutation p	[0.146]			[0.904]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,677,333	3,669,025	3,669,025	3,677,333	3,669,025	3,669,025
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966

Notes: Collapsed panel (product×destination×period cells, analytic weights), excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. Permutation p -values (treated-only design, $R = 1,000$) reported for export value only; columns (2)–(3) and (5)–(6) not permuted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

3 SUBSAMPLE C-PROD-H4

Table 3: C-prod-HS4 subsample — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0051)	−0.0019 (0.0043)	0.0005 (0.0007)	0.0005 (0.0011)	0.0001 (0.0010)	0.0002 (0.0004)
Asymptotic p	[0.858]	[0.665]	[0.471]	[0.680]	[0.906]	[0.689]
Bootstrap p	[0.860]	[0.662]	[0.421]	[0.730]	[0.916]	[0.713]
$EP_{dt} \times b_p$	0.0066 (0.0090)	0.0068 (0.0111)	−0.0017 (0.0070)	0.0024 (0.0066)	0.0033 (0.0059)	−0.0004 (0.0024)
Asymptotic p	[0.469]	[0.541]	[0.807]	[0.714]	[0.581]	[0.860]
Bootstrap p	[0.435]	[0.562]	[0.857]	[0.734]	[0.591]	[0.866]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,772,855	3,745,788	3,745,788	3,772,855	3,745,788	3,745,788
Clusters (d)	215	215	215	215	215	215
Adj. R^2	0.737	0.835	0.923	0.737	0.835	0.923
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0051)	−0.0032 (0.0042)	0.0012 (0.0012)	0.0004 (0.0013)	−0.0003 (0.0011)	0.0003 (0.0004)
Asymptotic p	[0.867]	[0.449]	[0.338]	[0.779]	[0.790]	[0.484]
Bootstrap p	[0.855]	[0.461]	[0.368]	[0.817]	[0.806]	[0.546]
$EP_{dt} \times b_p$	0.0033 (0.0168)	−0.0059 (0.0182)	0.0076 (0.0098)	0.0015 (0.0040)	−0.0006 (0.0032)	0.0022 (0.0025)
Asymptotic p	[0.846]	[0.747]	[0.440]	[0.719]	[0.854]	[0.383]
Bootstrap p	[0.850]	[0.758]	[0.554]	[0.735]	[0.852]	[0.454]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,772,321	3,745,254	3,745,254	3,772,321	3,745,254	3,745,254
Clusters (d)	214	214	214	214	214	214
Adj. R^2	0.737	0.835	0.923	0.737	0.835	0.923

Notes: C-prod-HS4 subsample: restricted to HS6 products with a comparable HS4-level match across classification revisions. Full firm-level panel, excluding Hong Kong and Macao. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

4 CEM DESTINATIONS

Table 4: CEM destinations — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0023 (0.0037)	−0.0014 (0.0041)	−0.0003 (0.0016)	−0.0006 (0.0011)	−0.0002 (0.0012)	−0.0004 (0.0003)
Asymptotic p	[0.543]	[0.734]	[0.837]	[0.576]	[0.857]	[0.201]
Bootstrap p	[0.610]	[0.824]	[0.887]	[0.609]	[0.878]	[0.241]
$EP_{dt} \times b_p$	−0.0041* (0.0024)	−0.0042 (0.0034)	−0.0002 (0.0013)	−0.0011 (0.0007)	−0.0006 (0.0010)	−0.0004 (0.0003)
Asymptotic p	[0.098]	[0.214]	[0.868]	[0.140]	[0.541]	[0.308]
Bootstrap p	[0.251]	[0.544]	[0.928]	[0.204]	[0.610]	[0.403]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	13,992,396	13,858,213	13,858,206	13,992,396	13,858,213	13,858,206
Clusters (d)	57	57	57	57	57	57
Adj. R^2	0.742	0.815	0.921	0.742	0.815	0.921
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0016 (0.0034)	−0.0016 (0.0035)	0.0000 (0.0019)	−0.0005 (0.0011)	−0.0003 (0.0011)	−0.0003 (0.0005)
Asymptotic p	[0.648]	[0.644]	[0.980]	[0.655]	[0.783]	[0.567]
Bootstrap p	[0.676]	[0.737]	[0.978]	[0.676]	[0.815]	[0.683]
$EP_{dt} \times b_p$	−0.0043* (0.0024)	−0.0055* (0.0030)	0.0001 (0.0013)	−0.0013* (0.0007)	−0.0011 (0.0009)	−0.0002 (0.0003)
Asymptotic p	[0.081]	[0.071]	[0.938]	[0.091]	[0.194]	[0.478]
Bootstrap p	[0.121]	[0.180]	[0.981]	[0.125]	[0.255]	[0.607]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	13,992,396	13,858,213	13,858,206	13,992,396	13,858,213	13,858,206
Clusters (d)	57	57	57	57	57	57
Adj. R^2	0.742	0.815	0.921	0.742	0.815	0.921

Notes: CEM-matched destinations subsample. Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). Quantity and unit-value regressions estimated on the subset with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

5 DEEP VS SHALLOW

Table 5: Deep vs. shallow agreements — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpt}^{val}$	(2) $\ln x_{fpt}^{qua}$	(3) $\ln x_{fpt}^{uv}$	(4) $\ln x_{fpt}^{val}$	(5) $\ln x_{fpt}^{qua}$	(6) $\ln x_{fpt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0022 (0.0035)	−0.0013 (0.0036)	−0.0004 (0.0011)	−0.0004 (0.0010)	−0.0001 (0.0010)	−0.0002 (0.0002)
Asymptotic p	[0.534]	[0.712]	[0.748]	[0.721]	[0.903]	[0.392]
Bootstrap p	[0.585]	[0.817]	[0.708]	[0.751]	[0.919]	[0.305]
$EP_{dt} \times b_p$	−0.0034 (0.0029)	−0.0034 (0.0034)	−0.0005 (0.0008)	−0.0006 (0.0010)	−0.0004 (0.0012)	−0.0003 (0.0003)
Asymptotic p	[0.246]	[0.330]	[0.572]	[0.549]	[0.712]	[0.389]
Bootstrap p	[0.499]	[0.601]	[0.771]	[0.739]	[0.785]	[0.379]
θ_{fpt}	✓	✓	✓	✓	✓	✓
θ_{fpt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	5,262,293	5,201,289	5,201,282	5,262,293	5,201,289	5,201,282
Clusters (d)	23	23	23	23	23	23
Adj. R^2	0.751	0.844	0.932	0.751	0.844	0.932
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0016 (0.0032)	−0.0015 (0.0031)	−0.0002 (0.0018)	−0.0003 (0.0010)	−0.0002 (0.0009)	−0.0002 (0.0004)
Asymptotic p	[0.619]	[0.633]	[0.915]	[0.776]	[0.832]	[0.624]
Bootstrap p	[0.616]	[0.727]	[0.910]	[0.784]	[0.857]	[0.715]
$EP_{dt} \times b_p$	−0.0042 (0.0025)	−0.0043 (0.0028)	−0.0007 (0.0009)	−0.0009 (0.0008)	−0.0008 (0.0009)	−0.0003 (0.0003)
Asymptotic p	[0.113]	[0.141]	[0.431]	[0.253]	[0.401]	[0.294]
Bootstrap p	[0.173]	[0.294]	[0.755]	[0.386]	[0.492]	[0.336]
θ_{fpt}	✓	✓	✓	✓	✓	✓
θ_{fpt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	5,260,451	5,199,451	5,199,444	5,260,451	5,199,451	5,199,444
Clusters (d)	23	23	23	23	23	23
Adj. R^2	0.751	0.844	0.932	0.751	0.844	0.932

Notes: Deep vs. shallow agreements subsample: restricted to destinations with at least one deep PTA in force. Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpt}^{uv}$). Quantity and unit-value regressions estimated on the subset with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

6 SATURATION LADDER

6.1 WB DEPTH

Table 6: Saturation ladder — WB EP depth (EP_{dt}^{WB})

	Baseline				Interaction (g_p, b_p)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A — $\ln x_{fpdt}^{val}$</i>								
EP_{dt}^{WB}	0.0031 (0.0025)	0.0046* (0.0027)	0.0001 (0.0033)	0.0009 (0.0022)				
$EP_{dt}^{WB} \times g_p$					0.0018 (0.0050)	0.0012 (0.0027)	0.0060 (0.0050)	0.0050 (0.0037)
$EP_{dt}^{WB} \times b_p$					−0.0037 (0.0150)	−0.0015 (0.0027)	0.0058* (0.0031)	0.0064** (0.0027)
Observations	29,477,365	35,607,036	22,927,402	29,473,145	29,477,365	35,607,036	22,927,402	29,473,145
Adj. R^2	0.795	0.621	0.859	0.798	0.795	0.621	0.859	0.798
<i>Panel B — $\ln x_{fpdt}^{qua}$</i>								
EP_{dt}^{WB}	0.0042 (0.0028)	0.0033 (0.0028)	−0.0003 (0.0034)	0.0012 (0.0023)				
$EP_{dt}^{WB} \times g_p$					0.0023 (0.0037)	0.0032 (0.0026)	0.0061 (0.0060)	0.0053 (0.0038)
$EP_{dt}^{WB} \times b_p$					−0.0025 (0.0131)	−0.0006 (0.0028)	0.0048 (0.0031)	0.0058** (0.0025)
Observations	29,259,694	35,355,735	22,748,753	29,255,449	29,259,694	35,355,735	22,748,753	29,255,449
Adj. R^2	0.871	0.749	0.911	0.873	0.871	0.749	0.911	0.873
<i>Panel C — $\ln x_{fpdt}^{uv}$</i>								
EP_{dt}^{WB}	−0.0010* (0.0005)	0.0014*** (0.0003)	0.0005*** (0.0002)	−0.0002 (0.0003)				
$EP_{dt}^{WB} \times g_p$					0.0001 (0.0016)	−0.0019 (0.0017)	0.0002 (0.0016)	0.0003 (0.0008)
$EP_{dt}^{WB} \times b_p$					−0.0012 (0.0026)	−0.0014** (0.0006)	0.0005 (0.0006)	0.0008 (0.0006)
Observations	29,259,686	35,355,728	22,748,748	29,255,441	29,259,686	35,355,728	22,748,748	29,255,441
Adj. R^2	0.956	0.932	0.973	0.957	0.956	0.932	0.973	0.957
θ_{fpd}	✓		✓	✓	✓		✓	✓
θ_{pt}				✓				✓
θ_{fpt}		✓	✓			✓	✓	
θ_{pd}		✓				✓		
θ_t	✓				✓			
Clusters (d)	231	232	231	231	231	232	231	231

Notes: Each column represents a different fixed-effects structure. Columns (1)–(4) estimate the baseline EP_{dt}^{WB} coefficient without product-type interactions. Columns (5)–(8) decompose the EP effect by interacting with environmental (g_p) and pollution-intensive (b_p) product classifications. Hong Kong and Macao included. All specifications include the non-environmental depth control (TD_{dt}) and its interactions (not shown). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Chinese Customs Office; World Bank DTA Database.

6.2 TREND DEPTH

Table 7: Saturation ladder — TREND EP count (EP_{dt}^{TR})

	Baseline				Interaction (g_p, b_p)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A — $\ln x_{fpdt}^{val}$</i>								
EP_{dt}^{TR}	0.0004 (0.0005)	0.0010** (0.0005)	0.0002 (0.0006)	0.0003 (0.0004)				
$EP_{dt}^{TR} \times g_p$					−0.0013 (0.0010)	0.0010 (0.0009)	0.0021** (0.0009)	0.0018** (0.0008)
$EP_{dt}^{TR} \times b_p$					−0.0049*** (0.0015)	0.000029 (0.0007)	0.0002 (0.0011)	0.0009 (0.0008)
Observations	29,477,365	35,607,036	22,927,402	29,473,145	29,477,365	35,607,036	22,927,402	29,473,145
Adj. R^2	0.795	0.621	0.859	0.798	0.795	0.621	0.859	0.798
<i>Panel B — $\ln x_{fpdt}^{qua}$</i>								
EP_{dt}^{TR}	0.0005 (0.0005)	0.0006 (0.0005)	0.000047 (0.0006)	0.0003 (0.0005)				
$EP_{dt}^{TR} \times g_p$					−0.0007 (0.0009)	0.0013 (0.0009)	0.0025** (0.0010)	0.0017** (0.0008)
$EP_{dt}^{TR} \times b_p$					−0.0034*** (0.0012)	0.0002 (0.0007)	0.0002 (0.0011)	0.0008 (0.0007)
Observations	29,259,694	35,355,735	22,748,753	29,255,449	29,259,694	35,355,735	22,748,753	29,255,449
Adj. R^2	0.871	0.749	0.911	0.873	0.871	0.749	0.911	0.873
<i>Panel C — $\ln x_{fpdt}^{uv}$</i>								
EP_{dt}^{TR}	−0.0001 (0.0001)	0.0004*** (0.0001)	0.0001*** (0.000047)	0.000022 (0.000048)				
$EP_{dt}^{TR} \times g_p$					−0.0004 (0.0003)	−0.0003 (0.0003)	−0.0002 (0.0003)	0.0002 (0.0002)
$EP_{dt}^{TR} \times b_p$					−0.0015*** (0.0003)	−0.0003* (0.0001)	−0.000005 (0.0001)	0.0002 (0.0001)
Observations	29,259,686	35,355,728	22,748,748	29,255,441	29,259,686	35,355,728	22,748,748	29,255,441
Adj. R^2	0.956	0.932	0.973	0.957	0.956	0.932	0.973	0.957
θ_{fpd}	✓		✓	✓	✓		✓	✓
θ_{pt}				✓				✓
θ_{fpt}		✓	✓			✓	✓	
θ_{pd}		✓				✓		
θ_t	✓				✓			
Clusters (d)	231	232	231	231	231	232	231	231

Notes: Each column represents a different fixed-effects structure. Columns (1)–(4) estimate the baseline EP_{dt}^{TR} coefficient without product-type interactions. Columns (5)–(8) decompose the EP effect by interacting with environmental (g_p) and pollution-intensive (b_p) product classifications. Hong Kong and Macao included. All specifications include the non-environmental depth control (TD_{dt}) and its interactions (not shown). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

7 SUB-INDEX DECOMPOSITION

7.1 FULL PANEL

Table 8: Full panel — sub-index decomposition

	WB sub-indices		TREND sub-indices				
	(1) Green Lib.	(2) Enf. DSM	(3) Green Mkt.	(4) Enf. DSM	(5) Hard	(6) Soft	(7) Reg. Space
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>							
$EP_{\text{sub}} \times g_p$	0.0085 (0.0467)	−0.0028 (0.0208)	0.0158 (0.0178)	0.0020 (0.0083)	0.0007 (0.0023)	−0.0037 (0.0085)	0.0049 (0.0095)
$EP_{\text{sub}} \times b_p$	−0.0019 (0.0335)	−0.0243** (0.0117)	−0.0088 (0.0153)	−0.0107* (0.0061)	−0.0019 (0.0016)	−0.0047 (0.0070)	0.0072 (0.0075)
θ_{fpd}	✓	✓	✓	✓	✓	✓	✓
$\theta_{f dt}$	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511
Clusters (d)	225	225	225	225	225	225	225
Adj. R^2	0.744	0.744	0.744	0.744	0.744	0.744	0.744
<i>Panel B — DESTA depth control</i>							
$EP_{\text{sub}} \times g_p$	0.0073 (0.0432)	−0.0029 (0.0168)	0.0141 (0.0162)	0.0016 (0.0087)	0.0005 (0.0023)	−0.0038 (0.0077)	0.0029 (0.0068)
$EP_{\text{sub}} \times b_p$	−0.0171 (0.0316)	−0.0308** (0.0150)	−0.0155 (0.0143)	−0.0133* (0.0068)	−0.0029 (0.0018)	−0.0080* (0.0043)	−0.0037 (0.0060)
θ_{fpd}	✓	✓	✓	✓	✓	✓	✓
$\theta_{f dt}$	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511
Clusters (d)	225	225	225	225	225	225	225
Adj. R^2	0.744	0.744	0.744	0.744	0.744	0.744	0.744

Notes: Full firm-level panel (excluding Hong Kong and Macao), restricted to observations with valid sub-index data. Each column estimates a separate model replacing the aggregate EP index with one sub-index; the non-environmental depth control ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) is included in all specifications but not shown. Fixed effects: firm-product-destination (θ_{fpd}), firm-destination-year ($\theta_{f dt}$), product-year (θ_{pt}). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

WB sub-indices: (1) Green Liberalization; (2) Enforcement & DSM. TREND sub-indices: (3) Green Market Access; (4) Enforcement & DSM; (5) Hard obligations; (6) Soft obligations; (7) Regulatory Space.

7.2 COLLAPSED PANEL

Table 9: Collapsed panel — sub-index decomposition

	WB sub-indices		TREND sub-indices				
	(1) Green Lib.	(2) Enf. DSM	(3) Green Mkt.	(4) Enf. DSM	(5) Hard	(6) Soft	(7) Reg. Space
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>							
$EP_{\text{sub}} \times g_p$	−0.0115 (0.0879)	−0.0048 (0.0394)	−0.0271 (0.0303)	0.0042 (0.0148)	0.0022 (0.0047)	0.0132 (0.0121)	0.0242** (0.0099)
$EP_{\text{sub}} \times b_p$	−0.0876* (0.0483)	0.0067 (0.0525)	−0.0491** (0.0243)	−0.0052 (0.0140)	−0.0013 (0.0037)	0.0019 (0.0108)	0.0225*** (0.0086)
θ_{pd}	✓	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023	3,681,023
Clusters (d)	228	228	228	228	228	228	228
<i>Panel B — DESTA depth control</i>							
$EP_{\text{sub}} \times g_p$	−0.0141 (0.0892)	−0.0105 (0.0280)	−0.0290 (0.0288)	0.0032 (0.0160)	0.0017 (0.0047)	0.0132 (0.0131)	0.0149 (0.0104)
$EP_{\text{sub}} \times b_p$	−0.1001** (0.0398)	−0.0197 (0.0343)	−0.0543*** (0.0187)	−0.0086 (0.0115)	−0.0023 (0.0029)	−0.0015 (0.0101)	0.0092 (0.0075)
θ_{pd}	✓	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333	3,677,333
Clusters (d)	228	228	228	228	228	228	228

Notes: Collapsed panel (HS6–destination–year cells), excluding Hong Kong and Macao. Dependent variable: mean log export value ($\bar{y}_{pdt}$), weighted by cell size. Each column estimates a separate model replacing the aggregate EP index with one sub-index; the non-environmental depth control ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) is included in all specifications but not shown. Fixed effects: product-destination (θ_{pd}), destination-year (θ_{dt}), product-year (θ_{pt}). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

WB sub-indices: (1) Green Liberalization; (2) Enforcement & DSM. TREND sub-indices: (3) Green Market Access; (4) Enforcement & DSM; (5) Hard obligations; (6) Soft obligations; (7) Regulatory Space.

8 PPML

Table 10: PPML — extensive margin (collapsed panel, zero-filled)

	WB EP depth (EP_{dt}^{WB})		TREND EP count (EP_{dt}^{TR})	
	(1)	(2)	(3)	(4)
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>				
$EP_{dt} \times g_p$	0.0015 (0.0046) [0.738]	0.0029 (0.0061) [0.630]	0.0001 (0.0019) [0.951]	−0.0001 (0.0021) [0.972]
Asymptotic p				
$EP_{dt} \times b_p$	−0.0301* (0.0158) [0.057]	−0.0268 (0.0178) [0.133]	0.0028 (0.0047) [0.550]	0.0048 (0.0050) [0.335]
Asymptotic p				
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,895,543	8,030,137	7,895,543	8,030,137
Clusters (d)	225	228	225	228
<i>Panel B — DESTA depth control</i>				
$EP_{dt} \times g_p$	0.0053 (0.0045) [0.232]	0.0064 (0.0050) [0.205]	0.0010 (0.0013) [0.442]	0.0008 (0.0015) [0.565]
Asymptotic p				
$EP_{dt} \times b_p$	−0.0186 (0.0169) [0.270]	−0.0108 (0.0198) [0.585]	0.0027 (0.0053) [0.612]	0.0046 (0.0050) [0.358]
Asymptotic p				
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,878,784	8,003,772	7,878,784	8,003,772
Clusters (d)	225	228	225	228

Notes: PPML estimated on the HS6–destination–year grid completed with zeros (Santos Silva & Tenreyro, 2006); fixed effects: product×destination (θ_{pd}), destination×year (θ_{dt}), product×year (θ_{pt}); standard errors clustered at the destination level. Columns (1) and (3) exclude Hong Kong and Macao; columns (2) and (4) include them. Panel A controls for non-environmental WB depth ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$, not shown); Panel B uses the DESTA depth index.

Log-linear estimates condition on positive flows: if an environmental clause created new green exports to a market where none previously existed, those estimates would miss it. PPML retains zeros and therefore captures this margin.

PPML coefficients are approximate percentage changes.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

9 DYNAMICS — EVENT STUDY (TWFE)

Table 11: Event study (TWFE) — EP leads and lags interacted with product type

Period	WB depth control		DESTA depth control	
	$EP \times g_p$ (1)	$EP \times b_p$ (2)	$EP \times g_p$ (3)	$EP \times b_p$ (4)
<i>Pre-treatment leads (parallel trends test)</i>				
$t = -6$	-0.044 (0.045)	0.027 (0.043)	-0.044 (0.045)	0.027 (0.043)
$t = -5$	-0.005 (0.038)	-0.017 (0.025)	-0.005 (0.038)	-0.017 (0.025)
$t = -4$	0.017 (0.027)	0.036 (0.027)	0.017 (0.027)	0.036 (0.027)
$t = -3$	-0.008 (0.020)	0.002 (0.028)	-0.008 (0.020)	0.002 (0.028)
$t = -2$	0.006 (0.019)	0.020 (0.039)	0.006 (0.019)	0.020 (0.039)
<i>Reference period</i>				
$t = -1$	0	0	0	0
<i>Post-treatment lags</i>				
$t = 0$	0.007 (0.014)	0.027 (0.027)	0.007 (0.014)	0.027 (0.027)
$t = 1$	0.010 (0.020)	0.017 (0.028)	0.010 (0.020)	0.017 (0.028)
$t = 2$	-0.020 (0.024)	0.021 (0.033)	-0.020 (0.024)	0.021 (0.033)
$t = 3$	-0.038 (0.026)	0.020 (0.034)	-0.038 (0.026)	0.020 (0.034)
$t = 4$	-0.039 (0.033)	0.021 (0.035)	-0.040 (0.033)	0.021 (0.035)
$t = 5$	-0.084** (0.034)	-0.032 (0.043)	-0.084** (0.034)	-0.032 (0.043)
Observations	3,681,023	3,681,023	3,677,333	3,677,333
Clusters (d)	228	228	228	228

Notes: TWFE event study on the collapsed panel. Each row corresponds to a relative time period t around PTA entry into force ($t = 0$). The dependent variable is log total exports from China. The coefficients are $EP_{dt} \times \mathbf{1}[\text{rel. time} = t] \times g_p$ (cols. 1, 3) and $EP_{dt} \times \mathbf{1}[\text{rel. time} = t] \times b_p$ (cols. 2, 4), estimated jointly. The non-environmental depth control ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) is included in all specifications but not shown. $t = -1$ is the omitted reference period. FE: $pd + dt + pt$. Standard errors clustered at the destination level in parentheses. Columns (1)–(2) use WB total depth; columns (3)–(4) use DESTA depth.

Pre-treatment coefficients ($t \leq -2$) are all statistically indistinguishable from zero, supporting the parallel-trends assumption. The negative coefficient on $EP \times g_p$ at $t = 5$ should be interpreted cautiously: the panel ends in 2015 and $t = 5$ covers the thinnest treated subsample.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic, destination-clustered).

10 LEAVE ONE OUT

Table 12: Leave-one-out: EP×dirty coefficient stability across 23 treated destinations

Excluded destination	WB depth control		DESTA depth control	
	Excl. HK/Macao (1)	Incl. HK/Macao (2)	Excl. HK/Macao (3)	Incl. HK/Macao (4)
<i>None (baseline)</i>	−0.0119***	−0.0189***	−0.0113***	−0.0082*
<i>None, extended dirty</i>	−0.0121***	−0.0191***	−0.0107***	−0.0071
<i>High EP depth only</i>	−0.0271**	−0.0456***	−0.0161	−0.0164
Bangladesh	−0.0118***	−0.0188***	−0.0113***	−0.0082*
Brunei	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Myanmar	−0.0118***	−0.0188***	−0.0112***	−0.0081*
Cambodia	−0.0119***	−0.0189***	−0.0112***	−0.0081*
India	−0.0120***	−0.0190***	−0.0115***	−0.0084**
Indonesia	−0.0119***	−0.0190***	−0.0119***	−0.0087**
Laos, PDR	−0.0119***	−0.0189***	−0.0113***	−0.0082*
Malaysia	−0.0124***	−0.0195***	−0.0110***	−0.0076*
Pakistan	−0.0114***	−0.0183***	−0.0106***	−0.0056
Philippines	−0.0118***	−0.0189***	−0.0115***	−0.0083*
Singapore	−0.0098**	−0.0166***	−0.0128***	−0.0100**
Korea Rep.	−0.0097*	−0.0225**	−0.0104	−0.0085
Sri Lanka	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Thailand	−0.0120***	−0.0189***	−0.0115***	−0.0082*
Vietnam	−0.0122***	−0.0192***	−0.0112***	−0.0080*
Timor-Leste	−0.0119***	−0.0189***		
Iceland	−0.0119***	−0.0189***	−0.0112***	−0.0081*
Switzerland	−0.0133***	−0.0209***	−0.0115***	−0.0084*
Chile	−0.0121***	−0.0191***	−0.0104***	−0.0071
Costa Rica	−0.0114***	−0.0185***	−0.0109***	−0.0076*
Peru	−0.0133***	−0.0204***	−0.0121***	−0.0092**
Australia	−0.0103	−0.0245**	−0.0110***	−0.0048
New Zealand	−0.0119***	−0.0190***	−0.0113***	−0.0081*
Hong Kong		−0.0129***		−0.0112***
Macau		−0.0183***		−0.0082*
Observations	3,681,023	3,769,187	3,677,333	3,759,733
Clusters (<i>d</i>)	228	230	228	230

Notes: OLS on the collapsed panel. Dependent variable: log total exports from China. The coefficient shown is $\hat{\beta}$ on EP×dirty, re-estimated dropping one treated destination at a time. FE: $pd + dt + pt$. Standard errors (not shown) clustered by destination. Columns (1) and (3) exclude Hong Kong and Macao from the sample; columns (2) and (4) include them as additional treated observations. Columns (1)–(2) use WB total depth as the control-depth variable; columns (3)–(4) use DESTA depth. Observations and clusters refer to the baseline specification; individual rows differ slightly due to the dropped destination.

High EP depth only: estimated on the subsample excluding Peru, Switzerland, and Korea jointly — the three destinations with above-median WB EP depth.

Timor-Leste rows are blank in columns (3)–(4): Timor-Leste is absent from the DESTA dataset. Hong Kong and Macau rows are blank in columns (1) and (3): those territories are already excluded in those samples.

Note on inference. Stars are based on asymptotic p -values with destination-clustered SEs. With ≈ 23 treated destinations, these p -values are substantially understated relative to the wild cluster bootstrap: the baseline coefficient in column (1) has asymptotic $p < 0.001$ but bootstrap $p = 0.072$ (Table 16). Stars are not comparable to the rest of the paper.

How to read. This is a test of *coefficient stability*, not significance. The relevant question is how much the point estimate moves when one destination is dropped, not how many stars it carries.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

11 APPENDIX: FULL PANEL INCL HK/MC

Table 13: Full panel including Hong Kong and Macao — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0036)	−0.0001 (0.0038)	−0.0004 (0.0013)	−0.0001 (0.0010)	−0.0001 (0.0011)	−0.0001 (0.0003)
Asymptotic p	[0.804]	[0.984]	[0.734]	[0.913]	[0.960]	[0.847]
Bootstrap p	[0.833]	—	—	[0.919]	—	—
$EP_{dt} \times b_p$	−0.0041 (0.0024)	−0.0036 (0.0031)	−0.0008 (0.0010)	−0.0010 (0.0006)	−0.0008 (0.0008)	−0.0001 (0.0004)
Asymptotic p	[0.094]	[0.236]	[0.451]	[0.112]	[0.321]	[0.836]
Bootstrap p	[0.176]	—	—	[0.142]	—	—
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	23,560,110	23,334,037	23,334,027	23,560,110	23,334,037	23,334,027
Clusters (d)	227	227	227	227	227	227
Adj. R^2	0.871	0.910	0.963	0.871	0.910	0.963
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0020 (0.0035)	−0.0016 (0.0032)	−0.0004 (0.0019)	−0.0004 (0.0009)	−0.0003 (0.0009)	−0.0001 (0.0004)
Asymptotic p	[0.572]	[0.617]	[0.824]	[0.675]	[0.716]	[0.723]
Bootstrap p	[0.611]	—	—	[0.703]	—	—
$EP_{dt} \times b_p$	−0.0057** (0.0027)	−0.0076** (0.0031)	0.0008 (0.0010)	−0.0014* (0.0007)	−0.0017* (0.0009)	0.0002 (0.0003)
Asymptotic p	[0.034]	[0.015]	[0.419]	[0.052]	[0.058]	[0.524]
Bootstrap p	[0.035]	—	—	[0.082]	—	—
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	23,381,204	23,155,803	23,155,793	23,381,204	23,155,803	23,155,793
Clusters (d)	227	227	227	227	227	227
Adj. R^2	0.871	0.911	0.963	0.871	0.911	0.963

Notes: Full firm-level panel, including Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv} = \ln x_{fpdt}^{val} - \ln x_{fpdt}^{qua}$). Quantity and unit-value regressions are estimated on the subset of observations with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets; “—” indicates bootstrap not available for this outcome. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

11.1 COLLAPSED PANEL INCL HK/MACAO

Table 14: Collapsed panel including Hong Kong and Macao — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})	TREND EP count (EP_{dt}^{TR})
	(1)	(2)
	$\ln x_{fpdt}^{val}$	$\ln x_{fpdt}^{val}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>		
$EP_{dt} \times g_p$	0.0017 (0.0091)	0.0018 (0.0018)
Asymptotic p	[0.849]	[0.337]
Bootstrap p	[0.825]	[0.414]
Permutation p	[0.897]	[0.468]
$EP_{dt} \times b_p$	-0.0189*** (0.0061)	-0.00001 (0.0015)
Asymptotic p	[0.002]	[0.995]
Bootstrap p	[0.005]	[0.995]
Permutation p	[0.146]	[0.996]
θ_{pd}	✓	✓
θ_{dt}	✓	✓
θ_{pt}	✓	✓
Observations	3,769,187	3,769,187
Clusters (d)	230	230
<i>Panel B — DESTA depth control</i>		
$EP_{dt} \times g_p$	-0.0067 (0.0048)	0.0002 (0.0020)
Asymptotic p	[0.157]	[0.931]
Bootstrap p	[0.484]	[0.949]
Permutation p	[0.465]	[0.929]
$EP_{dt} \times b_p$	-0.0082 (0.0043)	0.0008 (0.0015)
Asymptotic p	[0.055]	[0.575]
Bootstrap p	[0.195]	[0.635]
Permutation p	[0.384]	[0.788]
θ_{pd}	✓	✓
θ_{dt}	✓	✓
θ_{pt}	✓	✓
Observations	3,759,733	3,759,733
Clusters (d)	230	230

Notes: Collapsed panel (product×destination×period cells, analytic weights), including Hong Kong and Macao. Dependent variable: log export value ($\ln x_{fpdt}^{val}$). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. Permutation p -values (treated-only design, $R = 1,000$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

12 CO2 SHAPIRO

Table 15: CO₂ intensity (Shapiro 2021) — collapsed panel, triple-difference

	WB depth control (TD_{dt})		DESTA depth control	
	(1) Excl. HK/Macao	(2) Incl. HK/Macao	(3) Excl. HK/Macao	(4) Incl. HK/Macao
<i>Panel A — WB EP depth (EP_{dt}^{WB})</i>				
$EP_{dt} \times g_p$	−0.0050 (0.0070)	0.0004 (0.0080)	−0.0025 (0.0049)	−0.0041 (0.0055)
Asymptotic p	[0.472]	[0.956]	[0.613]	[0.455]
Bootstrap p	[0.544]	[0.961]	[0.797]	[0.797]
$EP_{dt} \times \text{CO}_2$	−0.0026 (0.0057)	−0.0049 (0.0064)	0.0013 (0.0031)	0.0028 (0.0031)
Asymptotic p	[0.653]	[0.448]	[0.679]	[0.373]
Bootstrap p	[0.707]	[0.529]	[0.773]	[0.384]
<i>Panel B — TREND EP count (EP_{dt}^{TR})</i>				
$EP_{dt} \times g_p$	0.0009 (0.0017)	0.0009 (0.0017)	0.0011 (0.0018)	−0.00002 (0.0017)
Asymptotic p	[0.605]	[0.617]	[0.546]	[0.990]
Bootstrap p	[0.677]	[0.698]	[0.652]	[0.993]
$EP_{dt} \times \text{CO}_2$	−0.0013** (0.0005)	−0.0014** (0.0005)	−0.0008** (0.0004)	−0.0002 (0.0006)
Asymptotic p	[0.011]	[0.012]	[0.032]	[0.779]
Bootstrap p	[0.063]	[0.056]	[0.083]	[0.817]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	3,681,023	3,769,187	3,677,333	3,759,733
Clusters (d)	228	230	228	230

Notes: Collapsed panel (product×destination×year cells, analytic weights). The binary dirty indicator is replaced by a continuous product-level CO₂ emission intensity from Shapiro (2021), mapped to HS6 codes. $EP_{dt} \times g_p$ captures the differential effect on environmental goods; $EP_{dt} \times \text{CO}_2$ captures the marginal effect of a one-unit increase in CO₂ intensity. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times \text{CO}_2$ controls (not shown). Columns (1) and (3) exclude Hong Kong and Macao; columns (2) and (4) include them. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database; Shapiro (2021).

13 WILD CLUSTER BOOTSTRAP p -VALUES

Table 16: Wild cluster bootstrap p -values (9,999 replications)

	(1)	(2)	(3)	(4)
<i>Panel A — collapsed panel (WB index)</i>				
EP depth $\times$ Green: coefficient	-0.00457	0.00173	-0.00433	-0.00675
Bootstrap p -value	0.649	0.825	0.546	0.484
EP depth $\times$ Dirty: coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Bootstrap p -value	0.072	0.005	0.047	0.195
<i>Panel B — collapsed panel (TREND index)</i>				
EP count $\times$ Green: coefficient	0.00181	0.00178	0.00161	0.00017
Bootstrap p -value	0.390	0.414	0.508	0.949
EP count $\times$ Dirty: coefficient	0.00035	-0.00001	-0.00025	0.00084
Bootstrap p -value	0.857	0.995	0.905	0.635
<i>Panel C — full firm-level panel (WB index)</i>				
EP depth $\times$ Green: coefficient	-0.00226	-0.00090	-0.00185	-0.00196
Bootstrap p -value	0.686	0.833	0.673	0.611
EP depth $\times$ Dirty: coefficient	-0.00435	-0.00409	-0.00559	-0.00574
Bootstrap p -value	0.185	0.176	0.049	0.035
<i>Panel D — full firm-level panel (TREND index)</i>				
EP count $\times$ Green: coefficient	-0.00010	-0.00011	-0.00012	-0.00039
Bootstrap p -value	0.931	0.919	0.920	0.703
EP count $\times$ Dirty: coefficient	-0.00089	-0.00101	-0.00140	-0.00140
Bootstrap p -value	0.177	0.142	0.069	0.082

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Asymptotic p -values from destination-clustered standard errors are unreliable with 23 treated clusters (roughly 14 distinct EP profiles). The wild cluster bootstrap reconstructs the coefficient distribution by resampling 9,999 times and produces substantially more conservative p -values. The baseline dirty coefficient (Panel A, column 1) has asymptotic $p < 0.001$ but bootstrap $p = 0.072$.

Panels C and D computed via `boottest` in Stata on residualized data (Frisch-Waugh), because the command does not support more than one set of absorbed fixed effects.

14 PERMUTATION TEST

Table 17: Permutation test: 1,000 random reassignments of environmental content

	(1)	(2)	(3)	(4)
<i>Panel A — WB index</i>				
EP depth $\times$ Green: observed coefficient	-0.00457	0.00173	-0.00433	-0.00675
Permutation p -value	0.597	0.897	0.480	0.465
Valid permutations	1,000	1,000	1,000	1,000
EP depth $\times$ Dirty: observed coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Permutation p -value	0.278	0.146	0.146	0.384
Valid permutations	1,000	1,000	1,000	1,000
<i>Panel B — TREND index</i>				
EP count $\times$ Green: observed coefficient	0.00181	0.00178	0.00161	0.00017
Permutation p -value	0.160	0.468	0.316	0.929
Valid permutations	1,000	1,000	1,000	1,000
EP count $\times$ Dirty: observed coefficient	0.00035	-0.00001	-0.00025	0.00084
Permutation p -value	0.817	0.996	0.904	0.788
Valid permutations	1,000	1,000	1,000	1,000

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

The EP profile of each agreement (depth level and entry years) is randomly reassigned among the 23 treated destinations, 1,000 times, and the model is re-estimated each time. The p -value is the share of random reassignments that produce a coefficient at least as large in absolute value as the observed one.

A high p -value means the same number would have arisen by randomly labeling these destinations — the result does not depend on actual environmental content. A low p -value means the observed estimate stands out in the placebo distribution.

The 13 distinct EP profiles (of which approximately 9 are effectively independent, since the 10 ASEAN destinations share a single agreement) make the permutation distribution discrete; p -values are not reproducible to the last digit across implementations.

15 BOOTSTRAP p -VALUE MATRIX (DIRTY COEFFICIENT)

Table 18: Summary: dirty-product coefficient across four specification variants, with common inference method

	(1)	(2)	(3)	(4)
Collapsed panel: coefficient	-0.01187	-0.01887	-0.01134	-0.00820
Bootstrap p -value	0.072	0.005	0.047	0.195
<i>Full panel</i> : coefficient	-0.00435	-0.00409	-0.00559	-0.00574
Bootstrap p -value	0.185	0.176	0.049	0.035

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Coefficient on the interaction between environmental depth (WB index) and dirty products. This is the only coefficient in the paper that moves: the one on green products is zero everywhere.

All p -values in this table come from the *wild cluster bootstrap*, so all eight cells are comparable.

How to read it. On the *full panel* the result depends on which depth control is used: with the World Bank control the p -value stays above 0.17; with the DESTA control it falls below 0.05. The two controls have different partial correlations with environmental depth net of fixed effects (0.959 and 0.891, respectively): the more a control overlaps with the variable of interest, the less independent variation remains and the larger the standard error.

16 SUN–ABRAHAM EVENT STUDY

Table 19: Sun–Abraham interaction-weighted event study: composition gap estimates

Years since entry	Green — neutral gap		Dirty — neutral gap	
	Coeff.	<i>p</i>	Coeff.	<i>p</i>
−10	−0.0500	0.419	−0.0356	0.598
−9	−0.0128	0.881	0.0641	0.424
−8	−0.0001	0.999	−0.0478	0.436
−7	−0.0392	0.422	−0.0186	0.561
−6	0.0027	0.959	0.0466	0.335
−5	0.0022	0.961	−0.0447	0.378
−4	0.0345	0.359	−0.0210	0.538
−3	0.0021	0.938	−0.0451	0.189
−2	0.0034	0.861	−0.0080	0.832
+0	0.0116	0.504	0.0579	0.107
+1	0.0191	0.470	0.0661	0.192
+2	−0.0049	0.881	0.0765	0.241
+3	−0.0244	0.501	0.0921	0.156
+4	−0.0187	0.711	0.0906	0.181
+5	−0.0233	0.661	0.1153	0.117
+6	0.0027	0.959	0.1264	0.165
+7	−0.0320	0.541	0.1083	0.223
+8	−0.0779	0.181	0.0983	0.235
Aggregated ATT	−0.0421	0.269	0.0729	0.276

Notes: Estimated via `eventstudyinteract` in Stata (Sun and Abraham, 2021). Dependent variable: destination-year composition gap (mean log exports of green or dirty products minus mean log exports of neutral products, within destination-year cells). Standard errors propagate estimation uncertainty in cohort shares, as prescribed by Sun and Abraham (2021); `fixest::sunab` in R treats the shares as known weights and yields narrower standard errors — the ratio reaches 3–4 in periods identified by many discordant cohorts. Coefficients agree to 15 significant figures across implementations. Periods outside the $[-10, +8]$ window are estimated (t down to -15 and up to $+13$) but not displayed; the endpoints shown are individual period coefficients, not accumulated bins. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

17 DESTINATION-SPECIFIC TRENDS AND PRE-TREND TESTS

Table 20: Destination-specific trends and pre-trend tests (collapsed panel)

	(1)	(2)	(3)	(4)
<i>Panel A — destination×product-type linear trends (WB index)</i>				
EP depth × Green	-0.00510 (0.00360) [0.494]	-0.00200 (0.00436) [0.808]	0.00084 (0.00434) [0.896]	-0.00006 (0.00441) [0.995]
EP depth × Dirty	-0.00825* (0.00423) [0.275]	-0.00927** (0.00414) [0.249]	-0.01030*** (0.00239) [0.086]	-0.00930*** (0.00204) [0.037]
<i>Panel A' — same specification, TREND index</i>				
EP count × Green	-0.00217*** (0.00062) [0.015]	-0.00214*** (0.00061) [0.018]	-0.00111** (0.00053) [0.218]	-0.00153*** (0.00049) [0.074]
EP count × Dirty	-0.00137 (0.00115) [0.355]	-0.00142 (0.00114) [0.339]	-0.00231*** (0.00066) [0.048]	-0.00164** (0.00064) [0.109]
<i>Panel B — pre-trend test (WB index)</i>				
Pre-agreement slope, Green	0.01732 (0.01295)	0.00402 (0.01678)	0.01719* (0.00997)	0.02218 (0.01346)
Bootstrap <i>p</i> -value	0.678	0.705	0.543	0.497
Pre-agreement slope, Dirty	0.06292 (0.04393)	0.06237 (0.03983)	0.07109** (0.03161)	0.07056** (0.03159)
Bootstrap <i>p</i> -value	0.610	0.515	0.416	0.410
<i>Panel B' — pre-trend test (TREND index)</i>				
Pre-agreement slope, Green	0.00728* (0.00438)	0.00707 (0.00441)	0.00775* (0.00428)	0.00944** (0.00389)
Bootstrap <i>p</i> -value	0.189	0.199	0.173	0.090
Pre-agreement slope, Dirty	0.01512** (0.00728)	0.01512** (0.00721)	0.01839*** (0.00596)	0.01703*** (0.00613)
Bootstrap <i>p</i> -value	0.293	0.301	0.164	0.243

Notes: Columns: (1) Excl. HK/Macao, TotalDepth control (baseline); (2) Incl. HK/Macao, TotalDepth control; (3) Excl. HK/Macao, DESTA control; (4) Incl. HK/Macao, DESTA control.

Stars are based on asymptotic *p*-values (destination-clustered standard errors). Wild cluster bootstrap *p*-values are reported separately in brackets (Panels A/A') or below each coefficient (Panels B/B').

Panels A/A'. The main specification's fixed effects do not absorb a shock specific to green products in a particular destination that grows over time — i.e., a country signing environmental clauses precisely as its demand for green goods is rising. Adding a linear trend for each destination interacted with product type absorbs exactly that dynamic.

TREND result in Panel A'. The TREND green coefficient is highly significant asymptotically ($p < 0.001$ in the baseline column) and survives the bootstrap ($p = 0.015$). Panel B' shows, however, that the green gap was already rising *before* the agreement entered into force (pre-slope +0.007, bootstrap $p = 0.189$), making a pre-existing trend a plausible explanation and weakening any causal reading.

Panels B/B'. The pre-agreement slope of the composition gap is estimated on pre-treatment years. A slope significantly different from zero would indicate that green and neutral products were already diverging before the agreement.

Standard errors in parentheses, *p*-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

18 FULL-PANEL ROBUSTNESS BATTERY

Table 21: Robustness on the full panel: sample, control, and dependent-variable variants

Variant	× Green	× Dirty	Observations	R^2
<i>Panel A — WB index</i>				
With additional controls (tariffs, concentration, antidumping)	-0.00023 (0.00296) [0.935]	-0.00600** (0.00259) [0.022]	19,286,638	0.8736
Excluding the 11 ASEAN destinations	-0.00252 (0.00360) [0.482]	-0.00438* (0.00223) [0.051]	19,236,090	0.8719
Including Hong Kong and Macao	-0.00089 (0.00360) [0.804]	-0.00409* (0.00243) [0.094]	23,560,110	0.8706
Common-support products only	-0.00225 (0.00392) [0.566]	-0.00435* (0.00223) [0.052]	21,519,197	0.8734
Green share in firm's export basket		-0.00010 (0.00015) [0.502]	13,345,132	0.8510
<i>Panel B — TREND index</i>				
With additional controls (tariffs, concentration, antidumping)	+0.00039 (0.00091) [0.669]	-0.00085 (0.00068) [0.209]	19,286,638	0.8736
Excluding the 11 ASEAN destinations	-0.00102 (0.00102) [0.318]	-0.00183** (0.00072) [0.012]	19,236,090	0.8719
Common-support products only	-0.00011 (0.00099) [0.924]	-0.00090 (0.00062) [0.153]	21,519,197	0.8734
PTA partners only, deep vs. shallow	-0.00036 (0.00100) [0.721]	-0.00061 (0.00101) [0.549]	5,262,293	0.8839
Green share in firm's export basket		-0.00006** (0.00003) [0.043]	13,345,132	0.8510

Notes: Baseline variant (excl. HK/Macao, TotalDepth control). Estimated with **reghdfe**; fixed effects and clustering as in the main specification, except where noted.

Common support: only products exported to both agreement and non-agreement destinations are retained. For the others, no observed comparison exists.

Green share in basket: the dependent variable changes. Instead of export value, it is the share of the firm's revenue from green products to that destination. Fixed effects: firm×destination and year.

Standard error in parentheses, p -value in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

19 APEC VS OECD GREEN DEFINITION

Table 22: Alternative environmental-good definition: OECD list vs. APEC list

	OECD list (used in this paper)	APEC list (54 codes)
EP depth (WB) $\times$ Green	-0.00457 (0.00696) [0.512]	0.00501 (0.01266) [0.693]
EP count (TREND) $\times$ Green	0.00181 (0.00182) [0.320]	0.00319 (0.00208) [0.126]

Notes: Baseline variant, collapsed panel.

What this tests. There is no single definition of “environmental good”. This paper uses the OECD list; here it is replaced with the APEC list, a much narrower set agreed politically in 2012. If the result depended on which products we call green, the two columns would diverge.

Standard error in parentheses, p -value in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

20 MINIMUM DETECTABLE EFFECTS

Table 23: How large an effect would the design need to detect it

Index	Margin	Asymptotic std. error	Minimum detectable effect (1 s.d.)	Bootstrap CI half-width (1 s.d.)	Bootstrap CI (per unit)
WB	Green	0.0070	4.64%	5.90%	[-1.77%, 3.19%]
WB	Dirty	0.0030	1.97%	2.40%	[-1.84%, 0.18%]
TREND	Green	0.0018	4.16%	3.62%	[-0.18%, 0.71%]
TREND	Dirty	0.0016	3.65%	3.53%	[-0.32%, 0.54%]

Notes: Estimation sample: collapsed panel, excl. HK/Macao variant. Standard deviation of regressors computed on the effective sample and weighted.

Why this table matters. A null result can mean two very different things: either the effect does not exist, or it exists but is too small for the data to distinguish it from noise. This table says which.

How to read it. The minimum detectable effect (column 4) is the threshold below which the design is uninformative, computed from the asymptotic method ($2.8 \times$ standard error). Column 5 reports the half-width of the wild cluster bootstrap confidence interval: this is not an 80%-power MDE (which would be $1.43\times$ larger), but the bootstrap margin of error. The informative bound for the precision estimate is the [bootstrap CI] column: on the green margin, it rules out effects larger than about 3% per provision at 95% confidence.

This yields a more precise statement of the result: not “we find no effect,” but “we can rule out effects larger than this threshold.”

21 BRANDI BENCHMARK

Table 24: Benchmark against Brandi et al. (2020): equivalent effect and comparison with the confidence intervals

Margin	Brandi effect (log points)	Asy. CI upper	WCB CI upper/coef.	Ratio
Green (liberal)	0.1570	0.0055	0.0355	1/4 (WCB)
Dirty (restrictive)	-0.0513	-0.0044 (point est.)	–	1/12